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Our Ref: P51672EP
Your Ref:

Dear Sirs,

Patent application no. 22751200.1 in Europe
AFL IG LLC
MULTI-FIBER CONNECTOR APPARATUS

In response to the Communication pursuant to Rules 161(1) and 162 EPC dated 13 February 2024, we enclose herewith amended pages 18 and 19 to replace those pages currently on file.

A copy of the previous pages with marked-up changes is enclosed for the Examiner's assistance.

Claim amendments

Claim 1 has been amended to define that the multi-fiber connector apparatus comprises a housing comprising a pair of first multi-fiber connectors operably coupled to a second multi-fiber connector by a plurality of optical fibers within the housing.

The first multi-fiber connector now comprises a twelve-way multi-fiber connector and the second multi-fiber connector now comprises a sixteen-way multi-fiber connector – see previous claims 2 and 7, and the specification as filed.

Claim 1 also defines that the plurality of fibers comprises a factor-of-eight optical fibers coupling the pair of first multi-fiber connectors to the second multi-fiber connector – see previous claim 3, for example.

Previous claims 2, 3, 5-8, 10, 11, 14-16 and 18-20 have been deleted and remaining claims renumbered consecutively.

Amended claim 4 (previous claim 9) defines a fiber optic network comprising a multi-fiber connector apparatus according to claim 1, and a plurality of fiber array units each operably coupled to respective first multi-fiber connectors at respective first interfaces.

GOOD IDEAS, WELL PROTECTED

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Reference signs have been added to the claims where appropriate.

Amendment of the description is deferred.

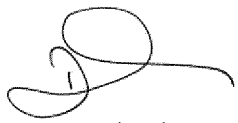
Novelty and inventive step

The combination of features now set out in amended claim 1 is novel and provides an inventive step over the cited prior art.

The claims now more precisely define the connection between a twelve-way multi-fiber connector and a sixteen-way multi-fiber connector by factor-of-eight fibers coupling therebetween.

The claimed multi-fiber connector apparatus may allow for improved transitioning from 400 Gb optical structures to 800 Gb, 1.6 Tb, or 3.2 Tb optical structures, such as may allow a datacentre operator to deploy a BASE16 structured cabling infrastructure when operating a 400 Gb structure (see paragraphs [0045]-[0046] of the specification as filed, for example). In contrast, D1 discloses first and second twelve-port connecters coupled to an equal number of ports across first, second and third eight-port connectors – see paragraph [0038], for example ('*The conversion device works with either universal routing or classic routing. It also works with n multiples of this configuration (n=1, 2, 3, ...), i.e., for n sets of two 12f connectors and n sets of three 8f connectors*'). The other cited prior art documents fail to teach or suggest an adapter interface that receives factor-of-eight fibers at a twelve-way multi-fiber connector interface and connects to a sixteen-way multi-fiber connector in any unequal quantity/ratio, such as to allow for a connection between 400 Gb optical structures with 800 Gb, 1.6 Tb or 3.2 Tb optical structures, for example.

Yours very truly,
SERJEANTS LLP



Darran Thacker
European Patent Attorney
(Association 13)

Encl:
Amended pages 18 and 19
Previous pages with marked-up changes